Fiscal Year:	FY 2022	**Task Last Updated:**	FY 01/02/2022
PI Name:	T'ien, James S. Ph.D.		
Project Title:	Flame Spread and Extinction Over Solid Materials (Growth and Extinction Limit, GEL)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jst2@case.edu	**Fax:**	FY 216 368-3007
PI Organization Type:	UNIVERSITY	**Phone:**	4408295283
Organization Name:	Case Western Reserve University		
PI Address 1:	Mechanical and Aerospace Engineering		
PI Address 2:	620 Glennan Hall, 10900 Euclid Avenue		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44106-1712	**Congressional District:**	11
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	03/01/2016	**End Date:**	02/28/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Olson, Sandra	**Contact Phone:**	216-433-2859
Contact Email:	Sandra.Olson@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 2/28/2023 per NSSC (Ed., 1/21/22) NOTE: End date changed to 2/28/2022 per PI (Ed., 1/22/21)		
Key Personnel Changes/Previous PI:	December 2020: Dr. Michael Johnston, Universities Space Research Association (USRA), is now a CoInvestigator.		
COI Name (Institution):	Ferkul, Paul Ph.D. (NASA Glenn Research Center/ National Center for Space Exploration Research) Olson, Sandra Ph.D. (NASA Glenn Research Center) Johnston, Michael C Ph.D. (Universities Space Research Association) Li, Chengyao (Case Western Reserve University)		
Grant/Contract No.:	NNX16AE84A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continuation of "Flame spread and extinction over solid materials," grant NNX10AD99A. Flammability of solid materials is of interest to NASA as a first line of defense in spacecraft fire safety efforts. In micro- or partial-gravity, the reduced buoyant flows, coupled with low-speed air circulation currents, create an environment quite different than that in normal Earth gravity. In addition, there are several oxidizer atmospheres under consideration for crew vehicles and habitats, which expand the scope of the problem. A fundamental understanding of solid flammability as a function of gravity, flow velocity, oxygen percentage, pressure, and sample configuration can be an important contribution both to combustion science and to the NASA space exploration initiatives. This proposed experiment will concentrate on the flame growth, decay, and extinction over the surface of a non-flat thick solid in microgravity. In particular, a solid sphere of substantial size (i.e., 4 cm diameter) is chosen as a representative of non-flat samples. In addition to the parameters influencing the flammability of thin solids, the degree of interior heat-up is an important parameter on the solid burning characteristics of thick specimens. For spherical samples, the degree of interior heating is always changing. The problem is therefore unsteady in nature. In addition, flow around a sphere is different from that around a flat surface. The existence of a forward stagnation point, and shoulder and wake regions results in a different local flow pattern, hence a different flame-solid interaction. These can affect the burning and extinction characteristics. In the proposed experiment, cast Polymethylmethacrylate (PMMA) spheres will be instrumented with five embedded thermocouples to record the interior temperatures during the preheating and the combustion processes. The project objectives are (1) Experimentally determine the flame growth characteristics (growth rate, flame shape, and dimensions) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; (2) Experimentally determine the flame extinction characteristics (quenching and blowoff limits) over a thick solid fuel as a function of flow velocity, oxygen percentage, pressure, and the degree of internal heating; and (3) Establish a high-fidelity numerical model that can be compared with the microgravity results and serve as a tool connecting normal gravity and microgravity performance. The proposed experiment is to be conducted aboard the International Space Station (ISS) in the Combustion Integrated Rack (CIR) because ground-based microgravity facilities cannot provide the test time duration needed for thick samples.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This is a fundamental study on the flammability of solids. The understanding to be achieved will help us to deal with fire safety on Earth and in space.
Task Progress:	1. Ignition Tests Properly controlled ignitions are critical to the success of the Growth and Extinction Limit (GEL) space experiment. Many parameters need to be quantified in the proposed ignition process using heated hot wires. A systematic experimental and modeling study has been carried out. Ignition limits and ignition delay times of a large polymethylmethacrylate (PMMA) sphere (4-cm diameter) and cylinder (5.08-m diameter) are investigated by placing an electrically-heated wire coil beneath the sample in a slow forced flow. The wire serves both as a heater for the solid sample and as a gas-phase hot spot to ignite the air-fuel vapor mixture. In the experiment, the wire electrical current and energizing duration, the wire placement distance from the sample surface, the forced flow velocity, oxygen percentage, and pressure are systematically varied. Ignition limits as a function of oxygen percentage, pressure, and igniter current are determined. Two ignition delay times are determined: Igniter-supported ignition and self-sustained ignition. These critical times are also functions of the degree of heat-up of the sample interior. Solid temperature profiles are measured using imbedded thermocouples to determine the sub-surface temperature gradients at the moment of ignition. The heat transfer modes from the igniter to the solid sample are investigated using a numerical analysis on the igniter and its interaction with the solid during the pre-pyrolysis heat-up period. At the flow stagnation-point region where the ignition is to occur, radiative flux is greater than convective flux. This is consistent with the experimental finding on ignition delay times and heat flux measurements. 2. Modeling Extinction Limits in Stagnation-point Diffusion Flame with Heat Loss to Solid Interior The planned microgravity experiment, Growth and Extinction Limit (GEL), is designed to study the material flammability and extinction of thermally thick solids with an emphasis to quantify the effect of conductive heat loss into the solid interior by measuring the in-depth temperature profile. To obtain a more detailed understanding of the extinction phenomena, a previously developed one-dimensional axisymmetric stagnation-point diffusion flame model is expanded to account for the excessive heat loss into the solid beyond steady burning state. This is important because for thick solids, the burning period can have a long and slow solid transient with a quasi-steady gas-phase. A non-dimensional excessive conductive heat loss parameter is defined in the paper, and extensive numerical computations have been performed to determine the extinction boundary and the characteristics of near-limit flames. The classical U-shaped flammability boundary in the oxygen vs. stretch rate plot now becomes a series of U-shaped boundaries as a function of this heat loss parameter. For a given oxygen ambient, extinction boundary is a function of the heat loss parameter. The boundary consists of a blowoff branch and a radiative quenching branch. Detailed flame structures and energy balances are discussed along the extinction boundary. In this model, the non-dimensional gas-phase reaction rate (ratio of residence time to reaction time) has the classical Damköhler number character. Along the extinction boundary, the maximum non-dimensional reaction rates are approximately constant and of the order unit across the entire range of stretch rates.
Bibliography Type:	Description: (Last Updated: 09/19/2025)
Abstracts for Journals and Proceedings	Li C, Johnston MC, T'ien J. "Extinction of the stagnation point diffusion flame: Effect of conductive heat loss into solid interior." 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. Abstracts. 37th Annual Meeting of the American Society for Gravitational and Space Research, Baltimore, MD, November 3-6, 2021. , Nov-2021